

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	04 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection	Import Electricity Consumption Dataset (CSV) Validate Dataset Structure
FR-2	Data Preparation	Clean Missing Values Format Data for Analysis
FR-3	Data Analysis	State-wise Consumption Analysis Region-wise Consumption Analysis Lockdown vs Recovery Analysis
FR-4	Dashboard & Visualization	Create Interactive Tableau Dashboards Apply Filters (State, Region, Year, Lockdown Phase) Generate Charts and Reports

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The Tableau dashboard should be easy to understand and allow users to interact using filters and visualizations.
NFR-2		

		dataset should be stored securely and protected from unauthorized modifications.
NFR-3	Reliability	The dashboard should consistently display accurate and error-free analysis results.
NFR-4	Performance	The system should load datasets quickly and generate visualizations with minimal delay.
NFR-5	Availability	The Tableau dashboard should be accessible whenever users need to analyze electricity consumption data
NFR-6	Scalability	The system should be capable of handling larger electricity consumption datasets, additional years, and more states or regions without significantly affecting dashboard performance.